Electricity Consumption Analysis Using Tableau

1. Introduction

Electricity consumption is an important indicator of economic activity and infrastructure development. Understanding electricity usage patterns helps governments and organizations plan better energy distribution and identify regions with high or low demand.

This project focuses on analyzing electricity consumption across different states and regions of India using Tableau. The goal is to identify trends, compare consumption patterns, and study the impact of lockdown on electricity usage.

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2. Problem Statement

The objective of this project is to analyze electricity consumption patterns across India and identify insights related to:

- Region-wise electricity consumption
- State-wise electricity usage
- Monthly and yearly trends
- Electricity consumption before and after lockdown

By visualizing the data using Tableau, we can better understand how electricity usage varies across different regions and time periods.

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3. Dataset Description

The dataset used in this project is **Consumption.csv** which contains electricity consumption data across various states in India.

The dataset contains the following fields:

- **State** – Name of the Indian state
- **Region** – Region classification (North, South, East, West, North East)
- **Latitude** – Latitude coordinate of the state
- **Longitude** – Longitude coordinate of the state
- **Date** – Date of electricity usage
- **Electricity Usage** – Amount of electricity consumed
- **Usage Category** – Category of usage (Low, Medium, High)

The dataset helps analyze electricity usage patterns geographically and temporally.

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4. Data Collection

The dataset was collected and stored in CSV format. This dataset contains multiple records representing electricity usage across states over different time periods.

The CSV dataset was imported into Tableau for further analysis and visualization.

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5. Data Preparation

Before creating visualizations, the dataset was prepared and verified in Tableau.

The following steps were performed:

1. Import the **Consumption.csv** dataset into Tableau.
2. Verify field types such as dates, numerical values, and categorical fields.
3. Ensure that state and region fields are correctly categorized.
4. Check for missing or incorrect values in the dataset.

After verifying the dataset, it was ready for visualization.

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6. Data Visualization

Multiple visualizations were created in Tableau to analyze electricity consumption patterns.

The following visualizations were created:

1. **2019 State Consumption** – Shows electricity usage by state for the year 2019.
2. **2020 State Consumption** – Displays electricity usage by state for the year 2020.
3. **Total Consumption** – Total electricity usage across states.
4. **Region Usage** – Electricity consumption grouped by regions.
5. **Top N and Bottom N States** – Highlights the states with the highest and lowest electricity consumption.
6. **Monthly Consumption (2019–2020)** – Displays the electricity usage trend across months.
7. **Region Wise Total Consumption** – Compares electricity usage across regions.
8. **Usage Before and After Lockdown** – Shows the impact of lockdown on electricity usage.
9. **Region Wise State Usage** – Displays state-level usage within each region.
10. **Quarter Usage** – Electricity consumption grouped by quarters.
11. **Metro City State Usage** – Electricity usage comparison among major cities.
12. **Yearly Usage** – Overall yearly electricity consumption trend.

These visualizations help analyze patterns and identify key insights.

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7. Dashboard

A dashboard was created in Tableau to combine multiple visualizations into a single interactive interface.

The dashboard contains:

- **Lockdown Impact Chart** – Comparison of electricity usage before and after lockdown.
- **Region Wise Consumption Chart** – Shows electricity usage by region.
- **Monthly Consumption Trend** – Displays electricity usage trends across months.

The dashboard allows users to quickly understand electricity consumption patterns.

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8. Story

A Tableau Story was created to present the analysis in a step-by-step narrative format.

The story explains:

- Electricity consumption patterns across regions
- Monthly consumption trends
- Electricity usage before and after lockdown

This storytelling approach helps present insights clearly and effectively.

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9. Publishing the Dashboard

After creating the dashboard and story, the project was published on **Tableau Public**.

Steps performed:

1. Sign in to Tableau Public.
2. Click **File → Save to Tableau Public**.
3. Provide the workbook title.
4. Publish the dashboard online.

This allows the dashboard to be shared publicly.

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10. Web Integration

The published Tableau dashboard was embedded into a web page using **HTML iframe**.

A simple HTML page was created to display the Tableau visualization within a website interface.

code:

```
<iframe
src="https://public.tableau.com/views/electricity_consumption_dashboard/Story1?:embed=yes&:showVizHome=no"
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width="1000"
height="700">
</iframe>
```

This integration allows users to interact with the Tableau dashboard directly on the webpage.

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11. Key Insights

From the analysis, the following insights were observed:

- Northern and Western regions show higher electricity consumption compared to other regions.
- Electricity usage fluctuates across months depending on demand and seasonal factors.
- Electricity consumption decreased significantly during the lockdown period.
- Some states consistently show higher electricity usage due to industrial and population factors.

These insights help understand energy consumption patterns across India.

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12. Conclusion

This project demonstrates how Tableau can be used to analyze and visualize electricity consumption data effectively.

By creating visualizations, dashboards, and stories, the project highlights important electricity usage trends and patterns across different regions and time periods.

The integration of Tableau with a web page also shows how data visualizations can be embedded into applications to provide interactive analytics.

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13. Tools Used

- Tableau Desktop Public Edition
- HTML
- CSV Dataset
- Web Browser